N_{2a} and N_{2b} (Group I, Row III): check of the split, UV handling, plus prescription

1 Check of the split $N_2 = N_{2a} + N_{2b}$

The colour factor of N_2 splits as

$$\begin{aligned} &- f^{adc} f^{fbe} U^{db}(x) U^{ce}(z) + N_c U^{af}(y) + N_c U^{af}(x) - N_c U^{af}(x) \\ &= \underbrace{\Big[- f^{adc} f^{fbe} U^{db}(x) U^{ce}(z) + N_c U^{af}(x) \Big]}_{N_{2a}} + \underbrace{\Big[N_c U^{af}(y) - N_c U^{af}(x) \Big]}_{N_{2b}}, \end{aligned} \tag{1.1}$$

and both parts keep the same kernel and prefactor, so the split is **correct**.

One typesetting slip: in your N_{2a} the brace $\Big\{$ is opened once but closed twice, and the colour factor stands outside the $\int_{x,y,z,x',w'}$ although it depends on x, y, z. The consistent form is

$$\begin{aligned} \frac{d^3 N_{2a\,\mathrm{NLO}}}{d^3 k} &= \frac{1}{(2\pi)^3} \frac{g^4}{8\pi^4} \frac{1}{k^+} \int_\Lambda^{k^+ - \Lambda} \frac{dp^+}{2\pi} \int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\, e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\, \frac{1}{[p^+ (y-x)^2 + (k^+ - p^+)\,(y-z)^2]} \\ &\times \frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2} \left[\frac{\delta^{im}\, d_\perp}{2\,(x-z)^2} \left[(y-x)^2 - (y-z)^2 \right] \frac{p^+\,(k^+ - p^+)}{k^{+\,2}} \right. \\ &- \frac{p^+}{(k^+ - p^+)} \left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2} - \frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2} \right] \\ &+ \frac{(k^+ - p^+)\,\delta_{im}}{k^+} \left[\frac{(y-z)\cdot(x-z)}{(x-z)^2} + \frac{(y-x)\cdot(y-z)}{2\,(y-x)^2} - \frac{\left[(y-x)^2 - (y-z)^2\right]}{2\,(x-z)^2} \right] \\ &- \frac{(k^+ - p^+)}{p^+} \left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2} + \frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2} \right] \\ &+ \frac{p^+\,\delta_{im}}{k^+} \left[\frac{(y-x)\cdot(x-z)}{(x-z)^2} - \frac{(y-x)\cdot(y-z)}{2\,(y-z)^2} - \frac{\left[(y-x)^2 - (y-z)^2\right]}{2\,(x-z)^2} \right] \\ &\left. - \left[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2} - \frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2} + \frac{(y-z)^m\,(x-z)^i}{(x-z)^2} + \frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2} \right] \right] \\ &\times \Big[U^{ab'}(x') - U^{ab'}(w') \Big] \Big[- f^{adc}\, f^{fbe}\, U^{db}(x)\, U^{ce}(z) + N_c\, U^{af}(x) \Big]\, \rho^{b'}(x')\, \rho^f(y). \end{aligned} \tag{1.2}$$

N_{2b} is exactly as you wrote it: the same expression with colour $\big[N_c U^{af}(y) - N_c U^{af}(x)\big]$.

2 Shorthand

$$\mathcal{N} = \frac{1}{(2\pi)^3} \frac{g^4}{8\pi^4} \frac{1}{k^+}, \qquad \frac{\mathcal{N}}{2\pi} = \frac{1}{(2\pi)^3} \frac{g^4}{16\pi^5} \frac{1}{k^+}, \qquad \xi = \frac{p^+}{k^+}, \qquad \lambda = \frac{\Lambda}{k^+}, \tag{2.1}$$

$$\Phi = e^{-ik\cdot(z-x)\frac{p^+}{k^+}}, \qquad D = p^+(y-x)^2 + (k^+ - p^+)(y-z)^2, \qquad T^{im} = \frac{(x'-w')^i (z-x)^m}{(x'-w')^2 (z-x)^2}, \tag{2.2}$$

$$B^{im} = \frac{(y-z)^i (x-z)^m}{(x-z)^2} + \frac{(y-x)^m (y-z)^i}{2(y-x)^2}, \qquad A^{im} = \frac{(y-x)^i (x-z)^m}{(x-z)^2} - \frac{(y-x)^i (y-z)^m}{2(y-z)^2}, \tag{2.3}$$

$$\mathcal{C}^{(a)} = \Big[U^{ab'}(x') - U^{ab'}(w') \Big] \Big[- f^{adc} f^{fbe} U^{db}(x) U^{ce}(z) + N_c U^{af}(x) \Big] \rho^{b'}(x') \rho^f(y), \tag{2.4}$$

$$\mathcal{C}^{(b)} = \Big[U^{ab'}(x') - U^{ab'}(w') \Big] \Big[N_c U^{af}(y) - N_c U^{af}(x) \Big] \rho^{b'}(x') \rho^f(y), \qquad \mathcal{C}^{(b)}_z = \mathcal{C}^{(b)}\big|_{U(x)\to U(z)}, \tag{2.5}$$

$$K_S = \frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2}\,\frac{(y-x)\cdot(y-z)}{(y-x)^2(x-z)^2}, \qquad P(\xi) = \xi(1-\xi) + \frac{\xi}{1-\xi} + \frac{1-\xi}{\xi}. \tag{2.6}$$

Part A: UV

3 Step 1: where the x integral can diverge

- **Large** x (for $0 < p^+ < k^+$): $T \sim 1/|x|$, $D \sim p^+x^2$, and the brackets are $O(1)$, so the integrand falls like $|x|^{-3}$. Finite.
- $x \to y$: the brackets contain at most $1/|y-x|$, which is integrable. At $p^+ \to k^+$, $D \to k^+(y-x)^2$, but $\int_0^{k^+} dp^+/D = \log[(y-z)^2/(y-x)^2]/[(y-z)^2-(y-x)^2]$ is only a logarithm, which is integrable. Finite.
- $x \to z$: terms of the form $1/(x-z)^2$, which diverge logarithmically in two dimensions. This is the **UV** region.

4 Step 2: the kernel as $x \to z$ (common to N_{2a} and N_{2b})

Put $r = x - z \to 0$, $n = y - z$, $a = x' - w'$; then $y - x = n - r$ and $(y-x)^2 - (y-z)^2 = -2n\cdot r + r^2$. At $r = 0$: $\Phi = 1$ and $D = k^+n^2$. Keep the $1/r^2$ terms and average over directions with $\langle r^i r^j\rangle = \frac{r^2}{2}\delta^{ij}$ (in d dimensions $\frac{r^2}{d}\delta^{ij}$ with $d_\perp = d$, which gives the same results):

$$d_\perp \text{ term:} \quad \frac{d_\perp}{2}\xi(1-\xi)\frac{(-a\cdot r)(-2n\cdot r)}{a^2r^4} \;\to\; \xi(1-\xi)\frac{a\cdot n}{a^2r^2}, \tag{4.1}$$

$$(1-\xi)\delta \text{ term:} \quad \text{bracket} = \frac{2n\cdot r}{r^2} + O(1) \;\Rightarrow\; -(1-\xi)\frac{a\cdot n}{a^2r^2}, \tag{4.2}$$

$$-\tfrac{1-\xi}{\xi}B \text{ term:} \quad T^{im}B^{im} = -\frac{a\cdot n}{a^2r^2} + O(1/r) \;\Rightarrow\; +\frac{1-\xi}{\xi}\frac{a\cdot n}{a^2r^2}, \tag{4.3}$$

$$\xi\delta \text{ term:} \quad \text{bracket} = \frac{2n\cdot r}{r^2} + O(1) \;\Rightarrow\; -\xi\frac{a\cdot n}{a^2r^2}, \tag{4.4}$$

$$\text{last bracket:} \quad \text{bracket} = -\frac{2n^m r^i}{r^2} + O(1) \;\Rightarrow\; +\frac{a\cdot n}{a^2r^2}, \tag{4.5}$$

$$-\tfrac{\xi}{1-\xi}A \text{ term:} \quad T^{im}A^{im} = -\frac{a\cdot n}{a^2r^2} + O(1/r) \;\Rightarrow\; +\frac{\xi}{1-\xi}\frac{a\cdot n}{a^2r^2}. \tag{4.6}$$

The constant terms cancel ($-(1-\xi) - \xi + 1 = 0$). With $1/D \to 1/(k^+n^2)$:

$$\text{kernel} \xrightarrow{x\to z} \frac{1}{k^+}P(\xi)\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2(x-z)^2} \;=\; \frac{1}{k^+}P(\xi)\,K_S\Big|_{x\to z}. \tag{4.7}$$

This was checked numerically: the angle-averaged kernel divided by K_S tends to $P(\xi)$ (4.41, 2.25, 9.2011 at $\xi = 0.2, 0.5, 0.9$), and the remainder falls like r^2.

5 Step 3: colour at $x \to z$

Adjoint Wilson lines leave f invariant, so $U^{db}(z)U^{ce}(z)f^{fbe} = f^{dcg}U^{gf}(z)$. Then

$$-f^{adc}f^{fbe}U^{db}(z)U^{ce}(z) + N_c U^{af}(z) = -f^{adc}f^{dcg}U^{gf}(z) + N_c U^{af}(z) = 0, \tag{5.1}$$

which was checked numerically for a random SU(3) matrix (residual 2×10^{-15}).

- N_{2a}**: UV finite.** $\mathcal{C}^{(a)}$ vanishes linearly as $x \to z$, so the integrand is at most $O(1/|x-z|)$, which is integrable. *No UV log and no subtraction are needed for* N_{2a}.
- N_{2b}**: UV divergent.** $\mathcal{C}^{(b)} \to \mathcal{C}^{(b)}_z \neq 0$, so the coefficient of the divergence is (4.7)$\times\mathcal{C}^{(b)}_z$.

6 Step 4: subtraction for N_{2b}

Define $S = \frac{1}{k^+}P(\xi)\,K_S\,\mathcal{C}^{(b)}_z$. It has exactly the behaviour (4.7) at $x \to z$, and it is integrable at $x \to y$ and at large x ($K_S \sim |x|^{-3}$). Write

$$N_{2b} = \underbrace{\mathcal{N}\int\frac{dp^+}{2\pi}\int e^{-ik\cdot(w'-z)}S}_{N_{2b\,\mathrm{UV}}} + \underbrace{\mathcal{N}\int\frac{dp^+}{2\pi}\int e^{-ik\cdot(w'-z)}\big[\text{integrand of } N_{2b} - S\big]}_{N_{2b\,\mathrm{finite}}}. \tag{6.1}$$

The second term is UV finite, so $d = 2$ ($d_\perp = 2$) can be used in it.

7 Step 5: the UV part $N_{2b\,\mathrm{UV}}$

5a. The x integral in two dimensions (shows where it diverges). Put $x = z + t$, $s = |t|$, $n = y - z$. Writing $n^2 - ns\cos\varphi = \frac{1}{2}[(n^2 + s^2 - 2ns\cos\varphi) + (n^2 - s^2)]$ and using $\int_0^{2\pi} \frac{d\varphi}{A - B\cos\varphi} = \frac{2\pi}{\sqrt{A^2 - B^2}}$:

$$\int_0^{2\pi} d\varphi\, \frac{n^2 - ns\cos\varphi}{n^2 + s^2 - 2ns\cos\varphi} = \pi + \pi\frac{n^2 - s^2}{|n^2 - s^2|} = \begin{cases} 2\pi & s < |n| \\ 0 & s > |n| \end{cases} \Rightarrow \int_x \frac{(y-x)\cdot(y-z)}{(y-x)^2(x-z)^2} = 2\pi\int_0^{|y-z|} \frac{ds}{s}, \tag{7.1}$$

which is divergent only at $s \to 0$, i.e. $x \to z$ (UV).

5b. In $d = 2 - \epsilon$. Introduce a Feynman parameter and shift $t \to t + \alpha n$:

$$\begin{aligned}\mu^{-\epsilon}\int d^d t\, \frac{(n-t)\cdot n}{(n-t)^2 t^2} &= \mu^{-\epsilon} n^2 \pi^{d/2}\Gamma\big(2 - \tfrac{d}{2}\big)\int_0^1 d\alpha\,(1-\alpha)\big[\alpha(1-\alpha)n^2\big]^{\frac{d}{2}-2} = \pi^{d/2}\Gamma\big(2-\tfrac{d}{2}\big)\frac{\Gamma(\frac{d}{2}-1)\Gamma(\frac{d}{2})}{\Gamma(d-1)}(\mu^2 n^2)^{-\frac{\epsilon}{2}} \\ &= \pi\left[\frac{2}{\epsilon_{\mathrm{UV}}} + 2\gamma_E + \log\frac{(y-z)^2\mu_{\overline{MS}}^2}{4}\right] + O(\epsilon), \qquad \mu_{\overline{MS}}^2 = 4\pi e^{-\gamma_E}\mu^2,\end{aligned} \tag{7.2}$$

which was checked with sympy.

5c. The p^+ integral, done exactly ($K_S\mathcal{C}_z^{(b)}$ does not depend on p^+):

$$\begin{aligned}\int_\lambda^{1-\lambda} d\xi\Big[\xi(1-\xi) + \Big(\frac{1}{1-\xi} - 1\Big) + \Big(\frac{1}{\xi} - 1\Big)\Big] &= \Big(\frac{1}{6} - \lambda^2 + \frac{2\lambda^3}{3}\Big) + 2\log\frac{1-\lambda}{\lambda} - 2(1-2\lambda) \\ &= 2\log\frac{k^+ - \Lambda}{\Lambda} - \frac{11}{6} + \frac{4\Lambda}{k^+} - \frac{\Lambda^2}{k^{+2}} + \frac{2\Lambda^3}{3k^{+3}}.\end{aligned} \tag{7.3}$$

5d. Result. The prefactor is $\frac{\mathcal{N}}{2\pi}\cdot\pi\cdot N_c$, so

$$\begin{aligned}\frac{d^3 N_{2b\,\mathrm{UV\,NLO}}}{d^3 k} &= \frac{1}{(2\pi)^3}\frac{g^4 N_c}{16\pi^4}\frac{1}{k^+}\left[2\log\Big(\frac{k^+ - \Lambda}{\Lambda}\Big) - \frac{11}{6} + \frac{4\Lambda}{k^+} - \frac{\Lambda^2}{k^{+2}} + \frac{2\Lambda^3}{3k^{+3}}\right]\int_{y,z,x',w'} e^{-ik\cdot(w'-z)} \\ &\times\left[\frac{2}{\epsilon} + 2\gamma_E + \log\frac{(y-z)^2\mu_{\overline{MS}}^2}{4}\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2}\Big[U^{ab'}(x') - U^{ab'}(w')\Big]\Big[U^{af}(y) - U^{af}(z)\Big]\rho^{b'}(x')\rho^f(y) \\ &\xrightarrow{\Lambda\to 0} \frac{1}{(2\pi)^3}\frac{g^4 N_c}{16\pi^4}\frac{1}{k^+}\left[2\log\frac{\vee}{\Lambda} - 2\log\frac{\vee}{k^+} - \frac{11}{6}\right]\int_{y,z,x',w'}(\cdots),\end{aligned} \tag{7.4}$$

using $2\log\frac{k^+}{\Lambda} = 2\log\frac{\vee}{\Lambda} - 2\log\frac{\vee}{k^+}$. This is the only place where $\log\mu$ appears.

Part B: plus prescription and the large longitudinal logs

8 Step 6: definitions

$$\Big[\frac{k^+}{p^+}\Big]_+ f(p^+) \equiv \frac{k^+}{p^+}\big[f(p^+) - f(0)\big], \qquad \Big[\frac{k^+}{q^+}\Big]_+ h(q^+) \equiv \frac{k^+}{q^+}\big[h(q^+) - h(0)\big],$$
$$\Big[\frac{k^+}{k^+ - p^+}\Big]_+ f(p^+) \equiv \frac{k^+}{k^+ - p^+}\big[f(p^+) - f(k^+)\big],$$

so that, exactly,

$$\int_\Lambda^{k^+-\Lambda} dp^+\frac{k^+}{p^+}f = \int_0^{k^+} dp^+\Big[\frac{k^+}{p^+}\Big]_+ f + k^+ f(0)\log\frac{k^+ - \Lambda}{\Lambda} + O(\Lambda). \tag{8.1}$$

The colour factors do not depend on p^+, so the same steps apply to N_{2a} and N_{2b}.

9 Step 7: the two singular terms of the kernel

7a. $p^+ \to 0$. Write $-\frac{k^+ - p^+}{p^+}B = -\frac{k^+}{p^+}B + B$ and $f = \Phi\, T^{im}B^{im}/D$. At $p^+ = 0$: $\Phi = 1$ and $D = k^+(y-z)^2$, so

$$T^{im}B^{im} = \frac{(x'-w')\cdot(y-z)\,(x-z)\cdot(z-x)}{(x'-w')^2(z-x)^2(x-z)^2} + \frac{(x'-w')\cdot(y-z)\,(y-x)\cdot(z-x)}{2(x'-w')^2(z-x)^2(y-x)^2}, \tag{9.1}$$

$$-k^+ f(0) = -\frac{T^{im}B^{im}}{(y-z)^2}. \tag{9.2}$$

7b. $p^+ \to k^+$**.** Set $p^+ = k^+ - q^+$; then $\int_\Lambda^{k^+-\Lambda} dp^+ = -\int_{k^+-\Lambda}^{\Lambda} dq^+$ and $-\frac{p^+}{k^+-p^+}A = -\big(\frac{k^+}{q^+}-1\big)A$, so

$$\int_{k^+-\Lambda}^{\Lambda} dq^+ \Big(\frac{k^+}{q^+}-1\Big)h = \int_{k^+}^{0} dq^+ \Big\{\Big[\frac{k^+}{q^+}\Big]_+ - 1\Big\}h - k^+h(0)\log\frac{k^+-\Lambda}{\Lambda}, \tag{9.3}$$

with $h = e^{-ik\cdot(z-x)\frac{k^+-q^+}{k^+}}T^{im}A^{im}/[(k^+-q^+)(y-x)^2+q^+(y-z)^2]$. At $q^+=0$ the denominator is $k^+(y-x)^2$, and the phase combines with $e^{-ik\cdot(w'-z)}$ into $e^{-ik\cdot(w'-x)}$:

$$T^{im}A^{im} = \frac{(x'-w')\cdot(y-x)\,(x-z)\cdot(z-x)}{(x'-w')^2(z-x)^2(x-z)^2} - \frac{(x'-w')\cdot(y-x)\,(y-z)\cdot(z-x)}{2(x'-w')^2(z-x)^2(y-z)^2}, \tag{9.4}$$

$$-k^+h(0) = -e^{-ik\cdot(z-x)}\frac{T^{im}A^{im}}{(y-x)^2}. \tag{9.5}$$

Both endpoint coefficients were checked numerically: for random points and $\Lambda/k^+ = 10^{-7}$, the Λ-cut integral agrees with the plus form plus the logs to 10^{-8}.

7c. The subtraction of N_{2b}**.** Write $P(\xi) = \xi(1-\xi)+\frac{k^+}{k^+-p^+}-1+\frac{k^+}{p^+}-1$. Each endpoint then gives $K_S\mathcal{C}_z^{(b)}\log\frac{k^+-\Lambda}{\Lambda}$:

$$-\frac{1}{k^+}\int_\Lambda^{k^+-\Lambda} dp^+ P\,K_S\mathcal{C}_z^{(b)} = -\frac{1}{k^+}\int_0^{k^+} dp^+\Big[\xi(1-\xi)+\Big[\frac{k^+}{k^+-p^+}\Big]_+ + \Big[\frac{k^+}{p^+}\Big]_+ - 2\Big]K_S\mathcal{C}_z^{(b)} - 2\log\frac{k^+-\Lambda}{\Lambda}K_S\mathcal{C}_z^{(b)}. \tag{9.6}$$

The first term is left unevaluated. Evaluated, it equals $+\frac{11}{6}K_S\mathcal{C}_z^{(b)}$.

10 Results for N_{2a}

N_{2a} has no UV part, so it splits into a finite part and a log part only.

$$\begin{aligned}
\frac{d^3N_{2a\,\text{finite NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\frac{g^4}{8\pi^4}\frac{1}{k^+}\int_0^{k^+}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\,\frac{1}{[p^+(y-x)^2+(k^+-p^+)(y-z)^2]}\\
&\times\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\Bigg[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\left[(y-x)^2-(y-z)^2\right]\frac{p^+(k^+-p^+)}{k^{+2}}\\
&+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2(y-x)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right]\\
&-\left[\frac{k^+}{p^+}\right]_+\left[\frac{(y-z)^i(x-z)^m}{(x-z)^2}+\frac{(y-x)^m(y-z)^i}{2(y-x)^2}\right]+\left[\frac{(y-z)^i(x-z)^m}{(x-z)^2}+\frac{(y-x)^m(y-z)^i}{2(y-x)^2}\right]\\
&+\frac{p^+\,\delta_{im}}{k^+}\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2(y-z)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right]\\
&-\left[\frac{(y-x)^m(x-z)^i}{(x-z)^2}-\frac{(y-x)^m(y-z)^i}{2(y-z)^2}+\frac{(y-z)^m(x-z)^i}{(x-z)^2}+\frac{(y-x)^i(y-z)^m}{2(y-x)^2}\right]\Bigg]\\
&\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[-f^{adc}f^{fbe}U^{db}(x)U^{ce}(z)+N_c\,U^{af}(x)\Big]\rho^{b'}(x')\,\rho^f(y)\\
&+\frac{1}{(2\pi)^3}\frac{g^4}{8\pi^4}\frac{1}{k^+}\int_{k^+}^0\frac{dq^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\frac{(k^+-q^+)}{k^+}}\,\frac{1}{[(k^+-q^+)(y-x)^2+q^+(y-z)^2]}\\
&\times\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\left[\frac{(y-x)^i(x-z)^m}{(x-z)^2}-\frac{(y-x)^i(y-z)^m}{2(y-z)^2}\right]\left\{\left[\frac{k^+}{q^+}\right]_+-1\right\}\\
&\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[-f^{adc}f^{fbe}U^{db}(x)U^{ce}(z)+N_c\,U^{af}(x)\Big]\rho^{b'}(x')\,\rho^f(y),
\end{aligned}\tag{10.1}$$

$$\begin{aligned}
\frac{d^3N_{2a\,\log\text{ NLO}}}{d^3k} &= -\frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}\log\Big[\frac{k^+-\Lambda}{\Lambda}\Big]\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\left[\frac{(y-z)\cdot(x'-w')\,(x-z)\cdot(z-x)}{(x-z)^2\,(x'-w')^2\,(z-x)^2\,(y-z)^2}\right.\\
&\left.+\frac{(y-x)\cdot(z-x)\,(y-z)\cdot(x'-w')}{2(y-x)^2\,(x'-w')^2\,(z-x)^2\,(y-z)^2}\right]\\
&\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[-f^{adc}f^{fbe}U^{db}(x)U^{ce}(z)+N_cU^{af}(x)\Big]\rho^{b'}(x')\rho^f(y)
\end{aligned}$$

$$
\begin{aligned}
&- \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\Big[\frac{k^+-\Lambda}{\Lambda}\Big]\int_{x,y,z,x',w'} e^{-ik\cdot(w'-x)}\Big[\frac{(y-x)\cdot(x'-w')\,(x-z)\cdot(z-x)}{(x-z)^2\,(x'-w')^2\,(z-x)^2\,(y-x)^2}\\
&\quad - \frac{(y-x)\cdot(x'-w')\,(y-z)\cdot(z-x)}{2(y-z)^2\,(x'-w')^2\,(z-x)^2\,(y-x)^2}\Big]\\
&\quad \times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[-f^{adc}f^{fbe}U^{db}(x)U^{ce}(z)+N_cU^{af}(x)\Big]\rho^{b'}(x')\rho^f(y). \qquad (10.2)
\end{aligned}
$$

Both are UV finite because of the colour factor (Step 3). Here $\log\frac{k^+-\Lambda}{\Lambda}\to\log\frac{\vee}{\Lambda}-\log\frac{\vee}{k^+}$.

11 Results for N_{2b}

$N_{2b}=N_{2b\,\mathrm{UV}}$ (7.4) $+N_{2b\,\mathrm{finite}}+N_{2b\,\log}$, with

$$
\begin{aligned}
\frac{d^3N_{2b\,\mathrm{finite\,NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_0^{k^+}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\Big\{e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\,\frac{1}{[p^+(y-x)^2+(k^+-p^+)(y-z)^2]}\\
&\quad\times\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\Big[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,[(y-x)^2-(y-z)^2]\,\frac{p^+(k^+-p^+)}{k^{+2}}\\
&\quad+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\Big[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2(y-x)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\Big]\\
&\quad-\Big[\frac{k^+}{p^+}\Big]_+\Big[\frac{(y-z)^i(x-z)^m}{(x-z)^2}+\frac{(y-x)^m(y-z)^i}{2(y-x)^2}\Big]+\Big[\frac{(y-z)^i(x-z)^m}{(x-z)^2}+\frac{(y-x)^m(y-z)^i}{2(y-x)^2}\Big]\\
&\quad+\frac{p^+\,\delta_{im}}{k^+}\Big[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2(y-z)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\Big]\\
&\quad-\Big[\frac{(y-x)^m(x-z)^i}{(x-z)^2}-\frac{(y-x)^m(y-z)^i}{2(y-z)^2}+\frac{(y-z)^m(x-z)^i}{(x-z)^2}+\frac{(y-x)^i(y-z)^m}{2(y-x)^2}\Big]\Big]\\
&\quad\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(x)\Big]\rho^{b'}(x')\,\rho^f(y)\\
&\quad-\frac{1}{k^+}\Big[\frac{p^+(k^+-p^+)}{k^{+2}}+\Big[\frac{k^+}{k^+-p^+}\Big]_++\Big[\frac{k^+}{p^+}\Big]_+-2\Big]\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{(y-x)\cdot(y-z)}{(y-x)^2\,(x-z)^2}\\
&\quad\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(z)\Big]\rho^{b'}(x')\,\rho^f(y)\Big\}\\
&\quad+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{k^+}^0\frac{dq^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\;e^{-ik\cdot(z-x)\frac{(k^+-q^+)}{k^+}}\,\frac{1}{[(k^+-q^+)(y-x)^2+q^+\,(y-z)^2]}\\
&\quad\times\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\,\Big[\frac{(y-x)^i(x-z)^m}{(x-z)^2}-\frac{(y-x)^i(y-z)^m}{2(y-z)^2}\Big]\,\Big\{\Big[\frac{k^+}{q^+}\Big]_+-1\Big\}\\
&\quad\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(x)\Big]\rho^{b'}(x')\,\rho^f(y), \qquad (11.1)
\end{aligned}
$$

$$
\begin{aligned}
\frac{d^3N_{2b\,\log\,\mathrm{NLO}}}{d^3k} &= -\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\Big[\frac{k^+-\Lambda}{\Lambda}\Big]\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\Big[\frac{(y-z)\cdot(x'-w')\,(x-z)\cdot(z-x)}{(x-z)^2\,(x'-w')^2\,(z-x)^2\,(y-z)^2}\\
&\quad+\frac{(y-x)\cdot(z-x)\,(y-z)\cdot(x'-w')}{2(y-x)^2\,(x'-w')^2\,(z-x)^2\,(y-z)^2}\Big]\\
&\quad\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_cU^{af}(y)-N_cU^{af}(x)\Big]\rho^{b'}(x')\rho^f(y)\\
&\quad-\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\Big[\frac{k^+-\Lambda}{\Lambda}\Big]\int_{x,y,z,x',w'} e^{-ik\cdot(w'-x)}\Big[\frac{(y-x)\cdot(x'-w')\,(x-z)\cdot(z-x)}{(x-z)^2\,(x'-w')^2\,(z-x)^2\,(y-x)^2}\\
&\quad-\frac{(y-x)\cdot(x'-w')\,(y-z)\cdot(z-x)}{2(y-z)^2\,(x'-w')^2\,(z-x)^2\,(y-x)^2}\Big]\\
&\quad\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_cU^{af}(y)-N_cU^{af}(x)\Big]\rho^{b'}(x')\rho^f(y)\\
&\quad-\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,2\log\Big[\frac{k^+-\Lambda}{\Lambda}\Big]\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2}\,\frac{(y-x)\cdot(y-z)}{(y-x)^2(x-z)^2}
\end{aligned}
$$

$$\times \Big[U^{ab'}(x') - U^{ab'}(w')\Big]\Big[N_c U^{af}(y) - N_c U^{af}(z)\Big]\rho^{b'}(x')\rho^{f}(y). \tag{11.2}$$

12 Summary and checks

1. **Bookkeeping.** $N_2 = N_{2a\,\text{finite}} + N_{2a\,\log} + N_{2b\,\text{UV}} + N_{2b\,\text{finite}} + N_{2b\,\log} + O(\Lambda/k^+)$. Nothing is counted twice: $+\int S$ is $N_{2b\,\text{UV}}$, and $-\int S$ is the subtraction line of (11.1) together with the last term of (11.2).

2. **UV logs.** The only $\log\mu$ is in $N_{2b\,\text{UV}}$, (7.4), with the factor $\big[2\log\frac{k^+}{\Lambda} - \frac{11}{6}\big]$. N_{2a} has none, by colour. $N_{2b\,\text{finite}}$ has none: at $x \to z$ its main terms give $\big[\xi(1-\xi) + [\frac{1}{1-\xi}]_+ + [\frac{1}{\xi}]_+ - 2\big]K_S\mathcal{C}_z^{(b)}$, which the subtraction line cancels at every p^+. $N_{2b\,\log}$ has none: at $x \to z$ its three terms go as $(+1+1-2)K_S\mathcal{C}_z^{(b)} = 0$.

3. **Where each piece goes.**

   - $N_{2a\,\text{finite}}$ and $N_{2b\,\text{finite}}$ go to the finite (Other terms) file.
   - $N_{2a\,\log}$ and $N_{2b\,\log}$ go to the JIMWLK file.
   - The $-\frac{11}{6}$ part of $N_{2b\,\text{UV}}$ goes to the Divergent file, where it combines with Row II's $\frac{11}{3}$.
   - The $2\log\frac{k^+}{\Lambda}$ part of $N_{2b\,\text{UV}}$ is, before the x integral, $+2\log\frac{k^+}{\Lambda}K_S\mathcal{C}_z^{(b)}$. It cancels the last term of (11.2); these are the $\pm 2N_c$ blocks of the JIMWLK file.

4. **Large x (IR, not UV).** At $p^+ = 0$ the first log term of each part falls only like $1/x^2$. This is large-distance behaviour, and it has to be checked in the sum of all $\log(k^+/\Lambda)$ terms of the JIMWLK file.